

Soneium ShibuyaToken Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Hans

December 23, 2024

Contents

1 About Cyfrin 2

2 About This Document 2

3 Disclaimer 2

4 Copyright and License 2

5 Audit Overview 3

6 Audit Scope 3

7 Risk Classification 3

8 Protocol Summary 4

9 Attestation 4

1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Startale on Soneium ShibuyaToken. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated December 23, 2024, was delivered to Startale, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

© 2026 Cyfrin Inc. All rights reserved. Material from Startale or third parties remains the property of its owners.

This summary may be shared only in complete and unmodified form, so that any copy can be checked against the hash in its attestation. Short quotations with attribution to Cyfrin are permitted, provided they do not misrepresent this summary's scope or results. Except as permitted by applicable law, no part of this summary may otherwise be copied, modified, adapted, sold or sublicensed, or used to create derivative works, without Cyfrin's prior written permission.

Cyfrin and the Cyfrin logo are trademarks of Cyfrin Inc. and may not be used to suggest that Cyfrin has reviewed or endorses anything outside this summary's scope, or to misrepresent its results.

5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Soneium ShibuyaToken code provided by Startale. The findings consist of 1 Low severity issue with the remainder being informational.

Summary

Project Name	Soneium ShibuyaToken
Repository	ccip-contracts-registration
Commit	b8491f385c80...
Fix Commit	129bc6a11eff...
Audit Timeline	Dec 12th - Dec 18th, 2024
Methods	Manual Review, Fuzzing

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	0	0	0
Medium Risk	0	0	0
Low Risk	1	1	0
Informational	5	4	1
Gas Optimizations	0	0	0
Total	6	5	1

6 Audit Scope

This security review analyzes the ShibuyaToken contract implemented in `contracts/Shibuya.sol`.

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

The ShibuyaToken contract is a key component in bridging ASTR (Astar Network's native token) from Astar L1 to Soneium L2. Soneium is a Layer 2 solution built on OP Stack technology, developed through a collaboration between Sony Group Corporation and Startale.

The bridging system consists of two main components:

1. **ASTR L1 to Soneium L2 bridge:** uses Chainlink's CCIP for cross-chain transfers, with wrap-native functionality and a single approval transaction.
2. **ShibuyaToken contract (Soneium L2):** an ERC20 token with minting and burning, which supports CCIP integration for L1-L2 bridging by adhering to CCIP's Cross-Chain Token Requirements, and enables further bridging to other Superchain L2s through SuperchainERC20 standard (ERC7802) compliance.

9 Attestation

Cyfrin attests that it performed a security review of the Soneium ShibuyaToken code provided by Startale.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at `https://github.com/Cyfrin/cyfrin-audit-reports`. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2024-12-23-cyfrin-soneium-shibuya-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Startale has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=4dbf5da72b1c25b63b6e6176bde9d283646735beef80e1d54d916f3b42da7ca6
F=2024-12-23-cyfrin-soneium-shibuya-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.